

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – VII

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Class:TYIT	Batch:1
Date of Assignment:04/08/20206	Date/Time of Submission:04/08/2026

a. Develop a PWA that works in offline mode using Service Worker.

Code :

Service-worker.js

```
1 self.addEventListener("install", e => {
2   e.waitUntil(
3     caches.open("v1").then(c =>
4       c.add("offline.html")
5     )
6   );
7 });
8
9 self.addEventListener("fetch", e => {
10  e.respondWith(
11    fetch(e.request).catch(() =>
12      caches.match("offline.html")
13    )
14  );
15 });
```

Manifest.json

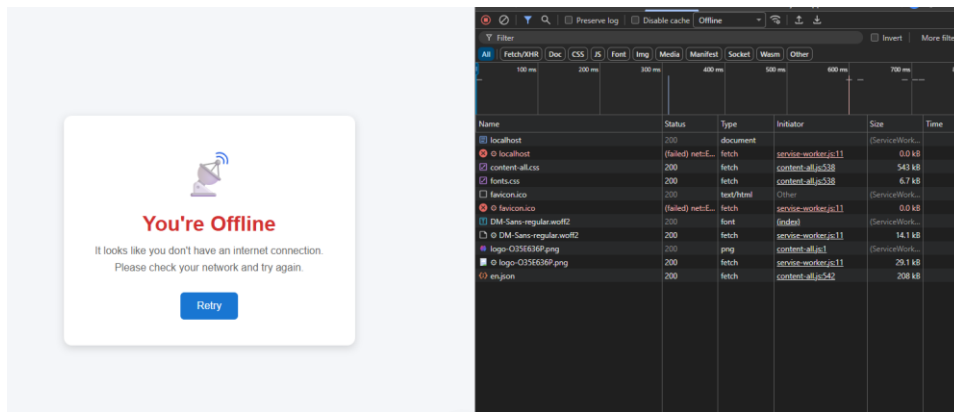
```
2  "name": "College Information Portal",
3  "short_name": "CIP",
4  "start_url": "/",
5  "display": "standalone",
6  "icons": [
7    {
8      "src": "/icons/icon1.png",
9      "sizes": "512x512",
10     "type": "image/png",
11     "purpose": "any"
```

layout.cs.html

```
<script>
  if ('serviceWorker' in navigator)
  {
    navigator.serviceWorker.register('/service-worker.js');
  }
</script>
<link rel="stylesheet" href="~/practical_7.styles.css" as="style">
<link rel="manifest" href="manifest.json"/>
</head>
```

Design Offline page :

Output :



b. Create a Student Record Management PWA.

Code :

```
const CACHE = "v1";

self.addEventListener("install", (e) => {
  e.waitUntil(
    caches.open(CACHE).then((cache) => {
      return cache.addAll([
        "/",
        "/Home/Index",
        "/css/site.css",
        "/js/site.js",
        "offline.html"
      ]);
    });
  );
});

self.addEventListener("fetch", (e) => {
  e.respondWith(
    caches.match(e.request).then((response) => {
      return response || fetch(e.request);
    })
  );
});
```

```
namespace WebApplication8.Models
{
    2 references
    public class student
    {
        1 reference
        public int RollNo { get; set; }
        1 reference
        public string Name { get; set; }
    }
}
```

```
@{
    ViewData["Title"] = "Home Page";
}

<div class="text-center">
    <h1 class="display-4">Welcome</h1>
    <p>Learn about <a href="https://learn.microsoft.com/aspnet/core">building Web apps with ASP.NET Core</a>.</p>
    <h1>Offline Page</h1>
    <h2>Student Record</h2>

    <p>Roll No: @Model.RollNo</p>
    <p>Name: @Model.Name</p>
</div>
```

```
7 {  
8 0 references  
9 public class HomeController : Controller  
10 {  
11 0 references  
12 public IActionResult Index()  
13 {  
14     student s = new student();  
15     s.RollNo = 47;  
16     s.Name = "Ashwin Tripathi";  
17     return View(s);  
18 }  
19 }
```

Output :

[practical_7](#) 

- [Home](#)
- [Privacy](#)

Welcome

Learn about [building Web apps with ASP.NET Core](#).

Offline Page

Student Record

Roll No: 47

Name: Ashwin Tripathi

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